

A Proposal of Acquiring and Analyzing Method for Distributed Litter on the Street using Smartphone Users as Passive Mobility Sensors

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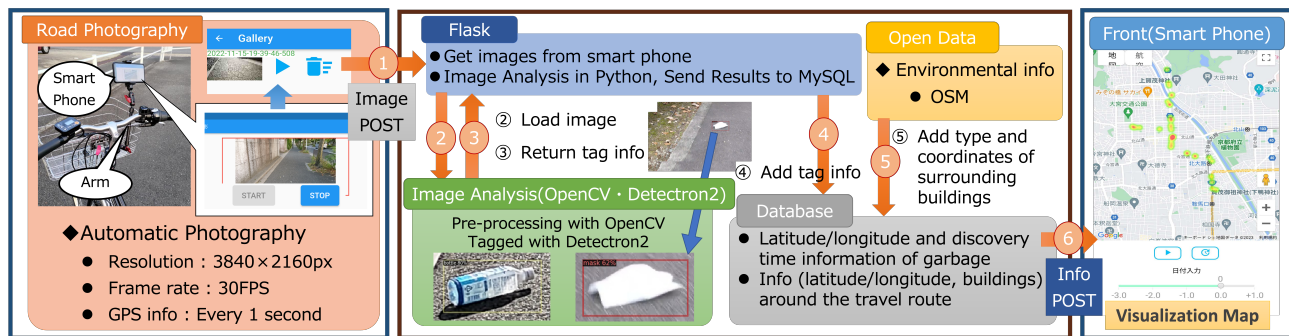


Figure 1: The litter acquiring and visualization system by the smartphone and bike as passive sensors

ABSTRACT

With increased environmental protection activities, smartphone-enabled cleaning activities to deter street littering are gaining attention. We propose a method to analyze litter-on-road images captured by a smartphone camera mounted on a bicycle for users who do not require conscious care (Fig.1). First, the user mounts the smartphone on a bicycle and starts the developed application, which creates a still image by capturing videos. The still images were then categorized using machine learning, and the type of trash was annotated in the images. Finally, to predict the distribution of trash, the probability of its influence on the environment, such as convenience stores and bars, was calculated using the machine learning model. This paper discusses our developed system's efficacy for acquiring and analyzing methods on the road. As a fast effort, we verify the accuracy of tagging PET bottles, cans, food trays, and masks using a learning model generated by Detectron2.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

Passive Mobility Sensors, mobile app, MaaS

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1 INTRODUCTION

As removing or preventing litter on the road, such as bottles, empty cans, and masks, becomes more critical, cleaning activities using smartphones to deter street littering are gaining attention. If people know about litter by visualizing online, rejuvenated cleaning activities and reduced littering through the psychological effect of the cleaned road could be expected. Sornlertlamvanich et al.[Sornlertlamvanich et al. 2022] shared information about litter by capturing photographs to visualize the pollution situation and improve environmental awareness. Sio et al.[Sio et al. 2022] created a

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system that automatically detects litter using YOLO and Raspberry Pi. It is essential to gather information for visualization; however, active cleaning users must carefully observe the road to capture images using a camera.

We propose a system for passively collecting litter-on-road images by conscious and unconscious mobility users, such as cyclists, to clean and solve these problems. Every cyclist included but not interested in clean-up activities, can easily and safely join the activity, and all mobile users can become passive sensors. This study verified the accuracy of tagging plastic bottles, cans, food trays, and masks using a learning model generated from Detectron2. It investigated the system's usefulness for acquiring and analyzing roadside environmental information. We also examine the system's effectiveness in acquiring and analyzing such information.

2 ANALYSIS SYSTEM OF LITTER ON THE ROAD BY A SMARTPHONE AS PASSIVE SENSORS

Figure 1 shows the flow of the analysis and visualization system for street litter. First, the user sets the smartphone on the bicycle and sends the captured video and its coordinates (latitude and longitude) to the server (Fig. (1)). After extracting still images from the video, the server performs preprocessing using OpenCV. The server then used machine learning to tag the objects (Figs. (2) and (3)) and store them in the database with the location information (Fig. (4)). In parallel, the server obtains the location information of stores (restaurants and convenience stores) associated with the trash from an Open Street Map (OSM) (Fig. (5)). Finally, the server predicts the distribution of trash based on the collected information (Fig. (6)). The predicted results were sent to the user's smartphone application to visualize the current and future trash distributions.

3 EVALUATION OF LITTER IMAGE ANALYSIS

Using Flutter and Python, we developed a smartphone application for cyclists to collect and annotate street litter images as a passive sensor. The smartphone was an Android device, and Detectron2[Wu et al. 2019] was the discriminative model. The Detectron2 Model Zoo provides a variety of pretrained models; we selected the Fast R-CNN R50-FPN. The TACO[Proença and Simões 2020] dataset was used as the training data for retraining the litter. In this study, four types of litter (plastic bottles, empty cans, food trays, and masks) were selected, labeled, and used as training data instead of the labeling provided in the dataset. Finally, a litter discrimination model was used to tag the images, and the results were stored in a database for each tag. We evaluated the litter discrimination model's accuracy and determined whether the litter was appropriately tagged as litter in the image as a percentage of recall and precision. We collected videos using a bicycle smartphone on campus roads under safe driving conditions for the evaluation. Images were not collected in the rain or while driving without light irradiation. During the verification, the average speed of the bicycles was 14.4 km/h, which was similar to the average speed of regular bicycles.

Table 1 shows the recall and precision of the annotations for all types. If the image annotated with the correct four types is the true data, TP is a True Positive, and TN is a True Negative. The total number was the sum of the analyzed still images. The

Table 1: Recall and Precision of Litter Annotations

Total#	TP	FN	FP	TN	Recall	Precision
2,260	164	165	36	1,895	0.498	0.820

Table 2: Recall rate of four litter types

PET bottle	can	food tray	mask
0.846	0.718	0.058	0.487

recall rate was only around 50%, but the precision rate was high at approximately 80%. Table 2 lists the detailed recall rates for the four types. The results showed that more than half of the masks and food trays could not be identified because both were similar in color and shape. In contrast, PET bottles and cans were better annotated. In the future, we will improve the model by acquiring more training data from a larger dataset and applying road condition data from OSM or SNS as features.

4 CONCLUSION

We proposed a safe and efficient roadside litter analysis system. Road images were acquired using a bicycle and a smartphone, and the image analysis effectiveness was examined. The results showed that while the precision rate was good, the recall rate was approximately 50%. In the future, we plan to enhance the accuracy of the discriminant model and develop a prediction model that considers related landmarks to visualize future litter distribution.

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